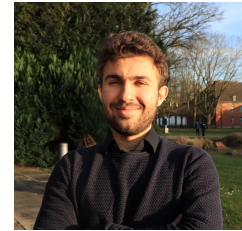


NIKOLAY L. TSONEV

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24.10.2004 · Birthplace: Sofia BG · GitHub · LinkedIn



EDUCATION

Institut Polytechnique de Paris, France M.Sc. in Parallel and Distributed Systems	09/2026 – 08/2028
Constructor University Bremen, Germany B.Sc. in Computer Science GPA: 1.86 (German scale)	09/2023 – 08/2026
International School of Riga, Latvia International Baccalaureate® (IB) Diploma Programme Score: 35/42	08/2021 – 08/2023

EXPERIENCE

Fraunhofer MEVIS <i>Thesis Researcher</i> - Investigating transformer internals by analysing concept features, attention pathways, and sparse MLP circuits via attribution graphs, activation interventions, and cross-layer transcoders. - Applying mechanistic interpretability to medical vision transformers to identify anatomy- and pathology-aligned feature directions; assessing explanation stability under domain shift.	Dec 2025 – June 2026 Bremen, Germany
Deutsches Zentrum für Luft- und Raumfahrt (DLR) <i>IT Working Student, High Performance Computing</i> - Maintain internal low-latency messaging software in C++17. - Developed an internal Python RabbitMQ API for broker communication and threshold-based topic monitoring. - Implemented tooling to detect and report missing entries in large scientific databases using SciPy. - Rewrote a legacy C messaging API as a modern non-blocking C++ interface, ensuring acknowledgment correctness under concurrent sender/receiver load using Bloomberg's thread-pool toolkit. - Developed Python APIs for RabbitMQ broker communication, including monitoring and alerting for topics falling below configurable thresholds. - Contributed to satellite data importers in Python.	June 2025 – May 2026 Munich, Germany
Constructor University Bremen <i>Teaching Assistant</i> <i>Programming in C/C++</i> (Fall 2024), <i>Algorithms & Data Structures</i> (Spring 2025), <i>Digital Systems & Computer Architecture</i> (Spring 2025) — led tutorials, marked assignments, and graded exams. <i>Operating Systems</i> (Fall 2025), <i>Databases</i> (Fall 2025), <i>Automata, Computability & Complexity</i> (Spring 2026). <i>Software Engineering</i> (Spring 2026) — selected by supervising professor to assist with simulator codebase extensions alongside four other TAs.	Sep 2024 – Present Bremen, Germany

PROJECTS

OML — Object Markup Language (GitHub) <i>Rust</i> - Cross-language data structure markup language: define a structure once, using the Object Markup Language, and transpile it to C++, Python, SQL, Kotlin, rust and other target languages; supports reverse compilation, converting data structures from the aforementioned languages back to OML and can be used to convert a DS from one language to another seamlessly. - Ships as a simple, user-friendly CLI.
PyFi — Quantitative Finance Library (GitHub) <i>C++20, CMake, pybind11, Python</i> - High-performance C++20 library with Python bindings for bond and option valuation. - Implements Black-Scholes (closed-form) and binomial tree pricing, yield curve construction, and Greeks computation.

Reproduce-Me — Scientific Code Reproducibility Toolkit (GitHub)

Python, LangGraph

- Agent-based pipeline that parses scientific papers and repositories to infer dependencies, runtime configurations, and execution environments; automates venv creation and demo synthesis.
- **1st place**, Constructor Knowledge Lab & Constructor Tech Hackathon.

ReciMe

Kotlin

- Android recipe app with a non-blocking UI built on coroutines; reduced ingredient list load time from over one minute to under ten seconds by parallelising API calls.
- Features grocery list management, ingredient filtering, and a home-screen “Meal of the Day” widget.

AudioMan

Kotlin

- Android audio player using coroutines for a non-blocking UI and a background playback service running outside the application lifecycle.
- Persistent storage of custom user audio and image data via serialisation.

TECHNICAL STACK

Languages	Python 3, C, C++17/20, TypeScript, Rust, SQL, Lean 4
Backend & APIs	REST, FastAPI, RabbitMQ
ML & Data	vLLM, LangChain/LangGraph, Pandas, NumPy, SciPy, Snowflake
Tools	CMake, pybind11, Git, Linux

AWARDS

1st Place — Constructor Tech × Constructor Knowledge Labs Hackathon
Top 5 — BMW Causal ML Hackathon
IBM Z Mainframe Certification

LANGUAGES

Bulgarian	Native
English	Bilingual
French	B2
German	B1
Russian	B1
Latvian	B1